Date: 20/06/2025

Lab Practical #03:

Study of different types of network cables & connectors and crimping a LAN.

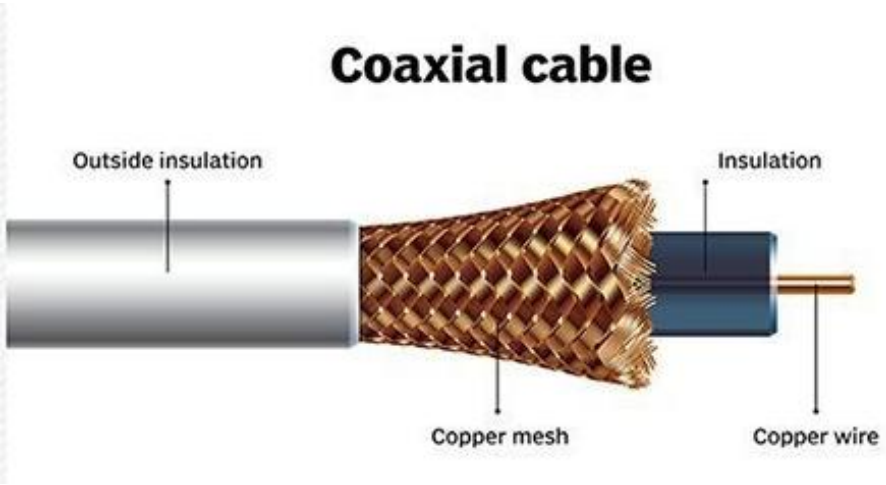
Practical Assignment #03:

1. List various networks cable. Also, write short description.
2. Difference between guided and unguided media.
3. Give cross-wired cable and straight through cable diagram (Color Code wise).

1. List various networks cable and connectors. Also, write short description.

a) Coaxial Cable :-

- **Description**:-
  - Coaxial cables have a central copper conductor surrounded by a dielectric insulator, a braided metal shield, and an outer jacket.
  - A coaxial cable (or coax cable) is a type of electrical cable used for transmitting data, video, and radio frequency (RF) signals. Its design allows it to carry high-frequency signals with minimal loss and interference.
  - The braided shield helps protect against electromagnetic interference (EMI) and noise.
- **Diagram**:



b) Twisted Pair Cable:-

- **Description**:-
  - These cables consist of pairs of insulated copper wires twisted together.
  - This twisting helps reduce electromagnetic interference.
  - Twisted pair cables are categorized into different specifications like Cat5e, Cat6, and Cat6a offering varying levels of speed and bandwidth.

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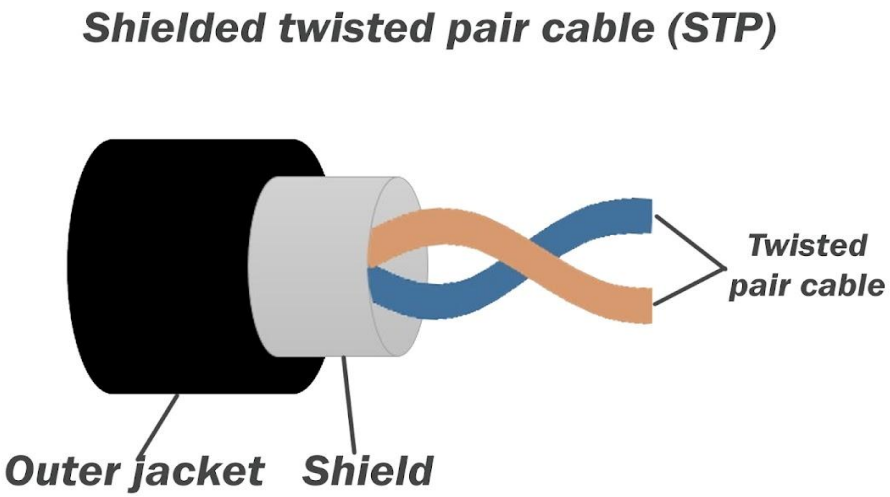
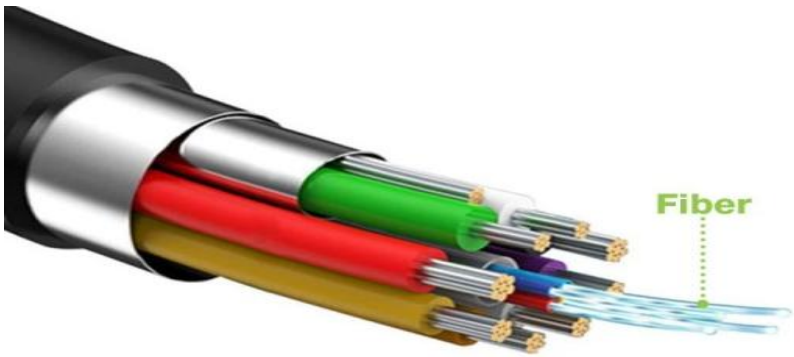
- **Diagram**:



C) Fiber Optic Cables :-

- **Description**:-
  - These cables use light signals to transmit data through a core of glass or plastic, offering high bandwidth and long-distance transmission capabilities.
  - They are used in high-speed networks and where electrical interference is a concern.
  - Common types: Single-mode and multi-mode fiber.

- **Diagram**:



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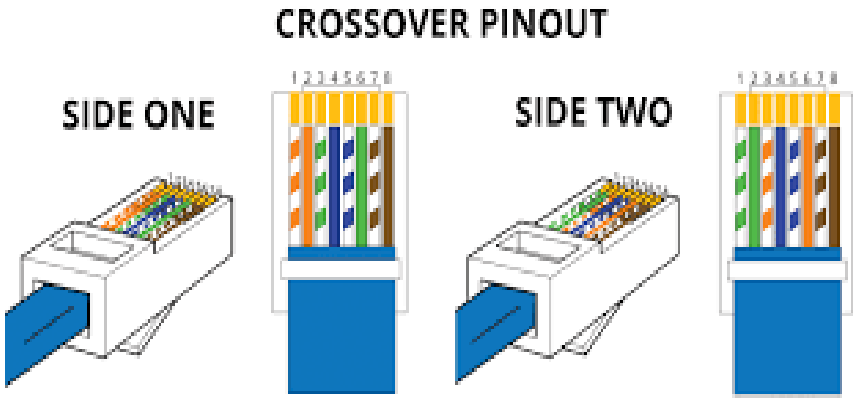
2. Difference between guided and unguided media.

Parameter	Guided Media	Unguided Media
Definition	Guided media uses physical cables (e.g., twisted pair, coaxial, fiber optic) to guide data signals from one point to another.	Unguided media transmits data through the air or space using electromagnetic waves (e.g., radio waves, microwaves, infrared).
Transmission Mode	Point-to-Point or Multipoint connections.	Broadcast, point-to-point, or point-to-multipoint connections.
Cost	Generally lower cost for short distances.	Higher cost for long distances (towers, satellite).
Example	Twisted Pair, Coaxial, Optical Fiber.	Radio Waves, Microwaves, Infrared, Satellite links.
Installation	Requires cables and connectors.	No cables required (uses antennas).

3. Give cross-wired cable and straight through cable diagram (Color Code wise).

a) Cross-wired Cable Diagram (Color Code)



b) Straight Through Cable Diagram (Color Code):-

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